

2(a)(i)	$E = \sigma / \varepsilon$ or $E = F / A\varepsilon$	C1
	$A = 1.4 \times 10^4 / (2.2 \times 10^{11} \times 0.0012)$ $= 5.3 \times 10^{-5} \text{ m}^2$	A1
2(a)(ii)	$(\Delta)h = 0.64 \times 0.49 (= 0.3136)$	C1
	$(\Delta)E = mg(\Delta)h$ or $W(\Delta)h$	C1
	$= 1.4 \times 10^4 \times 0.64 \times 0.49$ $= 4.4 \times 10^3 \text{ J}$	A1
2(b)	$P = Fv$ or W/t	C1
	$= (1.4 \times 10^4 \times 0.64) / 0.56$ or $(4.4 \times 10^3 / 0.49) / 0.56$	C1
	$= 1.6 \times 10^4 \text{ W}$	A1
2(c)	$m = 1.4 \times 10^4 / 9.81$ $(= 1427 \text{ kg})$	C1
	$(\text{resultant}) F = (1.4 \times 10^4 / 9.81) \times 1.3$ $(= 1855 \text{ N})$	C1
	$T = 1.4 \times 10^4 - 1855$ or $(1.4 \times 10^4 / 9.81) \times (9.81 - 1.3)$ $= 1.2 \times 10^4 \text{ N}$	A1
2(d)	upward sloping straight line from $(t_x, 0)$ to t_y	B1
	from t_y to t_z : an upward sloping curve with decreasing magnitude of gradient (that is horizontal at t_z)	B1